

Reinstalling an Older Version of MicroPython on ESP32

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This guide will walk you through reinstalling MicroPython v1.20 on an ESP32, covering both Windows and macOS. MicroPython v1.20 is recommended here as it retains support for the **hall sensor** function.

1. Download MicroPython v1.20 Firmware

- Visit the MicroPython ESP32 Downloads page.
https://micropython.org/download/ESP32_GENERIC/

Firmware

Releases

v1.24.0 (2024-10-25) .bin / [.app-bin] / [.elf] / [.map] / [Release notes] (latest)
v1.23.0 (2024-06-02) .bin / [.app-bin] / [.elf] / [.map] / [Release notes]
v1.22.2 (2024-02-22) .bin / [.app-bin] / [.elf] / [.map] / [Release notes]
v1.22.1 (2024-01-05) .bin / [.app-bin] / [.elf] / [.map] / [Release notes]
v1.22.0 (2023-12-27) .bin / [.app-bin] / [.elf] / [.map] / [Release notes]
v1.21.0 (2023-10-05) .bin / [.app-bin] / [.elf] / [.map] / [Release notes]
v1.20.0 (2023-04-26) .bin / [.elf] / [.map] / [Release notes]
v1.19.1 (2022-06-16) .bin / [.elf] / [.map] / [Release notes]
v1.18 (2022-01-17) .bin / [.elf] / [.map] / [Release notes]
v1.17 (2021-09-02) .bin / [.elf] / [.map] / [Release notes]
v1.16 (2021-06-23) .bin / [.elf] / [.map] / [Release notes]
v1.15 (2021-04-18) .bin / [.elf] / [.map] / [Release notes]
v1.14 (2021-02-02) .bin / [.elf] / [.map] / [Release notes]
v1.13 (2020-09-02) .bin / [.elf] / [.map] / [Release notes]
v1.12 (2019-12-20) .bin / [.elf] / [.map] / [Release notes]

Preview builds

v1.24.0-preview.476.g18c9abc32 (2024-10-24) .bin / [.app-bin] / [.elf] / [.map]
v1.24.0-preview.465.g6d98280fd (2024-10-24) .bin / [.app-bin] / [.elf] / [.map]
v1.24.0-preview.462.g078ead24f (2024-10-23) .bin / [.app-bin] / [.elf] / [.map]
v1.24.0-preview.461.g3f54e5dff (2024-10-22) .bin / [.app-bin] / [.elf] / [.map]
v1.23.0-5.g3613ad962 (2024-05-31) .bin / [.app-bin] / [.elf] / [.map]

(These are automatic builds of the development branch for the next release)

- Scroll to find **v1.20** in the **Firmware** section and download the **.bin** file that matches your ESP32 board.

2. Install esptool (Required for Flashing Firmware)

- Open a terminal (Command Prompt on Windows, Terminal on macOS) and install esptool by running:

```
pip install esptool
```

3. Find Your ESP32's Port

Windows:

- Open **Device Manager** (press Win + X and select Device Manager).
- Look under **Ports (COM & LPT)** for a port named something like **USB Serial Device (COMx)** or **Silicon Labs CP210x USB to UART Bridge (COMx)**.
- Note the COM port (e.g., COM3), which you'll use in the flash command.



USB-Enhanced-SERIAL CH9102 (COM3)

macOS:

- In **Terminal**, list all serial devices by running:

```
ls /dev/tty.*
```

- Look for a name like /dev/tty.usbserial-51850274651 or /dev/tty.SLAB_USBtoUART

and note it.

4. Optional: Erase Existing Firmware

- Erasing is optional but recommended for a fresh start.
- Run the command below, **replacing PORT** with your ESP32's port name:

For Windows:

```
esptool --chip esp32 --port PORT erase_flash
```

For Mac:

```
esptool.py --chip esp32 --port PORT erase_flash
```

- On Windows, replace PORT with COM3, COM4, etc.
- On macOS, replace PORT with the full path like /dev/ttyUSB0.

5. Flash MicroPython v1.20 Firmware

- For how to find the path of your firmware: right-click your firmware file, then copy, then paste that into your terminal, it will directly paste the path of your file
- Use this command to flash the downloaded .bin firmware:

for Windows:

```
esptool --chip esp32 --port PORT --baud 460800  
write_flash -z 0x1000 C:\Downloads\ESP32_GENERIC-  
20230426-v1.20.0.bin
```

for Mac:

```
esptool.py --chip esp32 --port /dev/tty.usbserial-51850274651 --baud 460800 write_flash -z 0x1000  
/Users/Downloads/ESP32_GENERIC-20230426-v1.20.0.bin
```

- Replace PORT path as in step 4, and specify the path to your .bin file for firmware.bin. The red ones are paths that your PORT path and file path should replace.

6. Verify the Installation

- Open Thonny, and see whether the default message shows

```
MicroPython v1.20.0 on 2023-04-26; ESP32 module with ESP32  
Type "help()" for more information.
```

Troubleshooting Tips

- **Port Not Recognized:** Ensure drivers are installed, especially on Windows (check the manufacturer's site for CP210x or CH340 drivers).
- **Permission Denied on macOS:** Run screen commands with sudo to avoid permission issues.